One Token per Trade: Multi-Resolution Limit Order Book Forecasting with a Foundation Model

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Abstract

Forecasting models for limit order books (LOB) are typically resolution-specific: top-of-book time series (L1), depth snapshots (L2), or event-stream prediction (L3). The atomic element of financial markets is the individual order or trade. We argue that an event-stream foundation model can produce forecasts at all three resolutions from one set of trained weights. Pairing with a market simulator – a deterministic world model of order-flow dynamics – enables auditable step-wise forecasts, essential for trading-desk deployment.

We train TradeFM, a 524M decoder-only Transformer, on over 10 billion order-flow tokens of US equity market data. We introduce a joint multivariate tokenization that encodes each trade event as one token factoring inter-arrival time, price level, volume, action, and side. Current time-series foundation models tokenize each feature independently and recover cross-feature structure through attention; we tokenize the joint event directly, so per-feature marginal forecasts reduce to slicing-and-summing over the joint softmax.

We evaluate at all three resolutions: (L3) event-stream 1-step NLL falls below the uniform baseline with zero-shot transfer across time and geography (US, CN, JP); (L2) recursive rollouts track emergent market factors – spreads, bid-ask volumes, order-book imbalance – more closely overall than Compound Hawkes and zero-intelligence baselines on Wasserstein and K-S distances; (L1) log-return distributions and canonical stylized facts hold across horizons (10s–120s). A single foundation-model + simulator architecture spans point and probabilistic forecasting, synthetic data generation, and counterfactual reasoning over market impact.

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1. Introduction

The atomic element of a financial market is the individual order or trade. Every coarser view – top-of-book prices (L1), depth snapshots (L2), the event stream itself (L3) – is a deterministic function of these events and a matching engine. Yet limit order book forecasting research has fragmented along these views: each resolution gets its own architectures and its own metrics (Zhang et al., 2019; Berti & Kasneci, 2025; Nagy et al., 2023; Wheeler & Varner, 2024; Li et al., 2024; Berti et al., 2025; Linna et al., 2025; Briola et al., 2024).

A foundation model trained on the joint event distribution forecasts every coarser view simultaneously: pair the model with a matching engine, and L1, L2, and L3 views follow from the sampled event stream. The data-processing inequality (Cover & Thomas, 2006, Thm. 2.8.1) formalizes this – the model's excess NLL on any coarser view is bounded by its excess NLL on the joint event distribution (Appendix E). One training objective controls error across all three resolutions.

TradeFM (Kawawa-Beaudan et al., 2026) is a 524M decoder-only Transformer trained from scratch on over 10 billion order-flow tokens of US equities; TFM-ITCH is a smaller 90M variant trained on 1.8 billion tokens of publicly available NASDAQ ITCH (Jan–Jun 2020). A joint multivariate tokenization encodes each event as a single token over a 16,384-element vocabulary factoring inter-arrival time, price level, volume, action, and side; per-feature marginal forecasts are slice-and-sum operations on the model's next-event distribution – no fitted parametric head, no rollout sampling. Current time-series foundation models tokenize each input variate independently (Ansari et al., 2024; Woo et al., 2024; Das et al., 2024); TradeFM tokenizes the joint event directly. TradeFM observes only the order-flow event stream (order submissions and cancellations) – the resting order book and true mid-price are never available as input – and the matching engine reconstructs L1, L2, and L3 views from each sampled event.

We evaluate at three resolutions: (L3) next-token perplexity on the event stream falls well below the uniform baseline and transfers zero-shot to APAC markets (CN, JP); (L2) recursive rollouts track emergent market factors – spreads, bid-ask volumes, order-book imbalance – more closely overall than Compound Hawkes and zero-intelligence baselines on Wasserstein and Kolmogorov-Smirnov distances across nine held-out months; (L1) log-return distributions match real data and reproduce canonical stylized facts across horizons (10s–120s) (Section 3).

Section 2 details the method, Section 3 reports multi-resolution evidence at L3/L2/L1, and Section 4 outlines extensions; data preprocessing, tokenizer mechanics, and reproducibility (Appendix H) are in the appendix.

2. Method

2.1. Trade Events as a Structured Sequence

We model market microstructure as a generative autoregressive problem over discrete trade events $E = (e_1, e_2, \ldots, e_T)$ (data sources and held-out splits in Appendix A). Each event $e_t = (\Delta t_t, \delta p_t, v_t, a_t, s_t)$ encodes inter-arrival time, price depth (basis points), volume (shares), action (add or cancel), and side (buy or sell).[1] Because the trade stream does not directly observe the order book, the mid-price is unobserved and must be reconstructed from executed trades. We use an **Exponentially-Weighted VWAP (EW-VWAP)** estimator, $\hat{p}_t^{\text{mid}} = \text{EMA}(p_t \cdot v_t) \,/\, \text{EMA}(v_t)$, which is comparable across heterogeneous liquidity profiles (Appendix A). Continuous features are mapped to scale-invariant, asset-agnostic ratios – log-volume, basis-point price depth relative to $\hat{p}^{\,\text{mid}}$, and intraday-relative price level versus the day's open – enforcing distributional stationarity across the breadth of US equities (Appendix A).

2.2. Joint Multivariate Tokenization

Each LOB event is encoded as one token from a 16,384-element vocabulary with per-feature radices $(n_{\Delta t}, n_{\delta p}, n_v, n_a, n_s) = (16, 16, 16, 2, 2)$, so $\prod_f n_f = 16{,}384$. The token ID factors cleanly back into its five features by integer division (Figure 1; encoding rules and worked example in Appendix B).

Per-feature marginal forecasts are closed-form slice-and-sum operations on the joint softmax:

$$P(\text{feature}_f = k \mid \text{ctx}) = \sum_{i:\, \text{digit}_f(i)=k} p_i(\text{ctx}) \qquad (1)$$

where $p_i(\text{ctx})$ is the joint softmax probability of token i. We minimize token-level negative log-likelihood – the strictly proper logarithmic scoring rule that underwrites automated market makers (Hanson, 2003) – with no fitted parametric head and no rollout sampling.

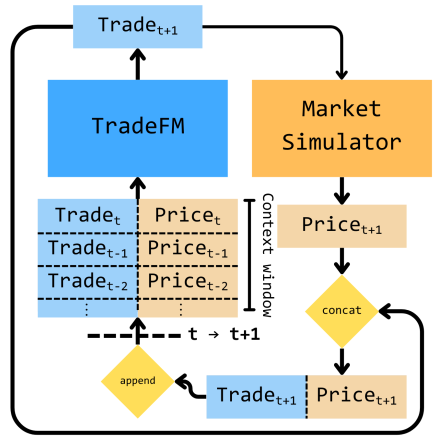

Figure 1. **TradeFM and the deterministic matching engine.** At each step, TradeFM proposes the next trade token Trade$_{t+1}$; the simulator consumes it and emits the updated Price$_{t+1}$. The concatenated (Trade$_{t+1}$, Price$_{t+1}$) is appended to the context and the loop continues. L1, L2, and L3 views are deterministic functions of the rolled-out token sequence.

This contrasts with current time-series foundation models. Chronos and TimesFM quantize each variate independently, with no cross-feature mechanism beyond per-variate attention (Ansari et al., 2024; Das et al., 2024); MOIRAI flattens per-variate patches into a unified sequence and recovers cross-feature structure through attention (Woo et al., 2024); Chronos-2 introduces multivariate forecasting via group attention (Ansari et al., 2025). The closest tokenization parallel is MusicGen's interleaved codebook over multiple audio streams (Copet et al., 2023), but interleaving imposes conditional independence between heads. We tokenize the joint event directly, preserving cross-feature dependence as a single softmax over $\prod_f n_f$ outcomes.

2.3. Architecture

TradeFM is a 524M decoder-only Transformer (Llama-style with grouped-query attention and rotary positional embeddings), trained from scratch on 10.7 billion order-flow tokens across more than 9,000 US equities. The full dataset spans 368 trading days (Feb 2024–Sep 2025); we reserve January 2025 onward as the held-out test set (Appendix A). The TFM-ITCH variant (90M parameters) is trained on 1.8 billion tokens of public NASDAQ ITCH (Jan–Jun 2020) and provides the cross-regime stress test in Section 3.3. Both variants share inputs and outputs: a 1,024-token context window over factored event tokens, a 16,384-element joint vocabulary, and a single softmax over the next-event distribution. Full configuration and training setup are in Appendix C.

[1] *Trade*, *event*, and *order* are used interchangeably for an order-flow message (submission or cancellation); a completed trade is a *fill* (execution).

2.4. Simulator as a Deterministic World Model

Real-world deployment of LOB forecasting models requires auditable, replayable rollouts – properties stochastic black-box models do not provide. We pair TradeFM with a deterministic LOB matching engine: a token sequence maps to L1, L2, and L3 views in closed form (Figure 1), and counterfactual interventions are bit-exact replayable, in the world-model spirit of Nagy et al. (2023) and Li et al. (2024). The engine is empirically high-fidelity, with fill-volume and fill-count correlations of 0.91 and 0.98 against real exchange replays (Appendix D).

3. Evaluation

We evaluate at three resolutions – L3 (Section 3.1), L2 (Section 3.2), L1 (Section 3.3) – and report distributional rather than point-forecast metrics. The motivation is operational: a desk pricing 30-second execution needs the distribution of mid-price moves, not just their average. Mid-price in liquid windows moves nearly randomly (Lo & MacKinlay, 1988; Cont, 2001), so a point forecast on the next-tick average is close to a random guess. What's tradeable is the shape of the distribution – its volatility, its asymmetry, its order-book imbalance – and the joint event distribution captures these directly (Bouchaud et al., 2018). Distributional fidelity is also the natural upstream metric for a foundation model: a task-specific point forecaster is fine-tuned from a well-calibrated joint, not the other way around.

Protocol. We evaluate TradeFM in three settings: 14,743 held-out US assets (Jan–Sep 2025) for L3 event-level perplexity; 9 stratified assets (across 3 liquidity tiers) $\times$ 9 held-out months $\times$ 10 rollouts per (asset, month) for L2 emergent-factor and L1 log-return rollouts; and CN + JP single-month holdouts (Jan 2025) for zero-shot APAC transfer. TFM-ITCH (90M, trained on public NASDAQ ITCH Jan–Jun 2020, 1.8B tokens) is tested on the 2025 holdout for cross-regime stress. Baselines are Compound Hawkes (Bacry et al., 2015) and zero-intelligence (Smith et al., 2003); metrics are token-weighted mean NLL for L3 (in the model's native token space) and Kolmogorov-Smirnov (D_n) and 1-Wasserstein (W_1) distances on per-feature marginal distributions for L2 and L1, aligned with the LOB-Bench protocol (Nagy et al., 2025).

3.1. L3: Event-stream Forecasting

Figure 2 shows per-region perplexity distributions over held-out batches.

Geographic Transfer. TradeFM's mean NLL on US held-out events (Jan–Sep 2025) is 2.81 nats – well below the closed-form uniform-vocabulary bound of $\ln(16{,}384) = 9.70$ nats, a $3.45\times$ compression. Tested

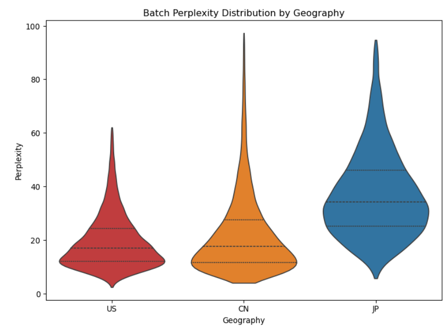

Figure 2. **Zero-shot geographic generalization.** Perplexity distributions for TradeFM (trained on US equities) evaluated on held-out US, China, and Japan data (January 2025) demonstrate cross-geography robustness of scale-invariant features.

zero-shot on China and Japan held-out batches (Jan 2025), perplexity degrades by less than 25%, placing APAC NLL below 3.03 nats – still well under uniform. This holds despite real structural differences in how these markets clear: Japan's Itayose batch auction, China's $\pm 10\%$ daily price limits, and wider Asian spreads (3–10 bps versus 1–2 bps for US large-cap). Same model, structurally distinct markets, no fine-tune – generalization across geography.

3.2. L2: LOB Snapshots and Emergent Factors

Table 1 reports K-S (Kolmogorov-Smirnov) and W_1 (Wasserstein) distances on six emergent order-flow factors (spreads, inter-arrival times, price depths, OBI, bid volume, ask volume) over nine held-out months of US equities.

Distributional Fidelity. TradeFM has the lowest Wasserstein distance on five of the six features; Hawkes is lowest on spreads. On K-S it is closest on four (inter-arrival times, price depths, OBI, ask volume); Hawkes is closest on two (spreads, bid volume). Hawkes is parametric, and its functional form concentrates probability near the modal feature value, where K-S – sensitive to the steepest part of the CDF – picks up its tighter fit. TradeFM is non-parametric over a 16,384-token vocabulary; it assigns probability wherever the training data has mass, including the rare large-move outcomes that Wasserstein rewards. Per-month decomposition of K-S and W_1 is in Appendix F (Figure 7); the aggregate result is not driven by one lucky month. Nine months past the training window, the joint event prior still tracks emergent LOB structure.

3.3. L1: Top-of-book Log-Returns and Stylized Facts

Log-Return Distributional Fidelity. L1 is the most decision-relevant resolution for short-term execution. Log-returns emerge from recursive rollouts: TradeFM samples each next event, the simulator updates the LOB state and mid-price, which is fed back to the model (Figure 1; Appendix A).

Quantity	K-S			Wasserstein		
	TFM	Hawkes	ZI	TFM	Hawkes	ZI
Spreads	0.238	**0.218**	0.400	0.400	**0.302**	0.375
IA Times	**0.281**	0.515	0.651	**0.318**	0.626	0.415
Price Depth	**0.169**	0.281	0.436	**0.339**	0.348	0.390
OBI	**0.142**	0.155	0.237	**0.099**	0.165	0.200
Bid Vol	0.386	**0.296**	0.460	**0.130**	0.278	0.616
Ask Vol	**0.360**	0.380	0.391	**0.160**	0.198	0.638

Table 1. **Order-flow distributional fidelity on 9 held-out months.** Lowest distance per row in bold. TFM (TradeFM) is closest on Wasserstein for five of six quantities (all but spreads) and on K-S for four; see Section 3.2 for the model-fit interpretation.

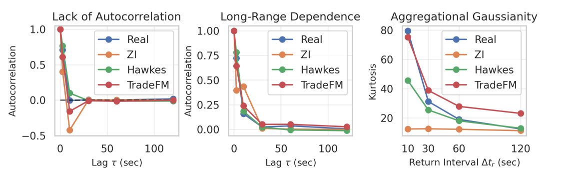

Figure 3. **Stylized facts of log returns.** TradeFM rollouts (red) versus real data (blue) over 9 assets $\times$ 9 held-out months. (Left) near-zero autocorrelation of returns. (Middle) slowly-decaying autocorrelation of $|r_{t,\Delta t}|$ (volatility clustering). (Right) heavy tails at short Δt aggregating toward Gaussian at longer horizons (aggregational Gaussianity).

Across 9 assets, 3 liquidity tiers, and 9 held-out months (Jan–Sep 2025), TradeFM achieves K-S distances $\{0.013, 0.024, 0.035, 0.043\}$ at horizons $\Delta t = \{10, 30, 60, 120\}$ s – 2–3$\times$ tighter than Compound Hawkes – and Wasserstein 5–6$\times$ tighter at every horizon (Figure 4; ECDF view in Figure 8 and per-horizon detail in Appendix F). Rollouts reproduce the canonical stylized facts of returns (Cont, 2001): near-zero autocorrelation of $r_{t,\Delta t}$ at lag $\tau > 1$, slowly-decaying autocorrelation of $|r_{t,\Delta t}|$ (volatility clustering), and heavy tails at short Δt aggregating toward Gaussian at longer horizons (Figure 3).

Cross-Regime Stress. TFM-ITCH (90M, public-data variant, trained on NASDAQ ITCH Jan–Jun 2020, 1.8B tokens) tested on the Jan–Sep 2025 holdout beats Compound Hawkes on log-return Wasserstein at every horizon by 21–27% (Table 3; details in Appendix F) while trailing on K-S – the same model-fit pattern as Section 3.2 (TradeFM concentrates mass on rare large moves; the parametric baseline concentrates near the modal frequency).

4. Discussion and Future Work

Limitations. We measure compression-grade quality (NLL, W_1, D_n) rather than calibration-grade quality; Property E.1 bounds KL on any coarser view derived from the joint events but does not transport calibration – randomized PIT (Czado et al., 2009) on the joint is the natural next experiment. L1 evaluation uses mid-price reconstructed via EW-VWAP (Appendix A); we report one rollout per context, not

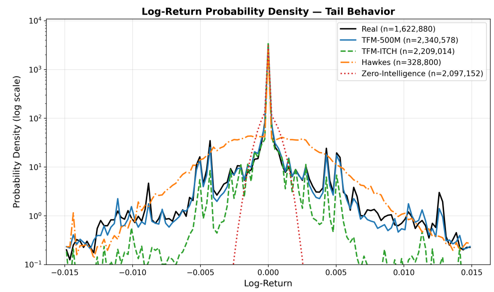

Figure 4. **Log-return probability density (log scale).** TradeFM and TFM-ITCH match the real density into the tails. Hawkes overestimates probability in the middle range; zero-intelligence has too narrow a peak and effectively zero probability outside it.

Monte-Carlo CRPS. We do not benchmark against Chronos, MOIRAI, or TimesFM (Ansari et al., 2024; Woo et al., 2024; Das et al., 2024): per-variate-tokenized TSFMs underperform domain-pretrained models on finance (Rahimikia et al., 2025), and adapting their architectures or tokenization schemes to event-stream LOB data is future work. The TFM-ITCH cross-regime result conflates a five-year temporal gap with a source change (public ITCH vs proprietary flow); the axes are not isolated. Closed-loop evaluation couples model and simulator quality; the zero-intelligence control through the identical simulator fails to reproduce stylized facts, Table 1's event-level marginals (Δt, δp, v) are scored pre-simulator, and real-order replays validate the engine (Appendix D) – full disentanglement is future work.

Vocabulary scaling and continuous tokenization. Our 16,384-element vocabulary is coarse beside modern time-series tokenizers (50–512 bins per variate; Ansari et al. 2024). Three paths preserve the joint design: hierarchical bucketing (RQ-VAE-style coarse-to-fine) keeps closed-form marginals at the bucket level; continuous per-factor heads (mixture-of-Gaussians, Student-t, or diffusion; Berti et al. 2025) trade closed-form marginals for direct continuous forecasts; hybrid tokenization – discrete action and side, continuous $(\Delta t, \delta p, v)$ – mirrors Linna et al. (2025).

Auxiliary-loss multi-task fine-tuning. Mid-price-change Δp_t is currently a conditioning input only. An auxiliary head with loss $\mathcal{L} = \lambda\,\mathrm{NLL}_{\text{event}} + (1-\lambda)\,\mathrm{NLL}_{\Delta p}$ biases representations toward L1 utility while preserving the joint objective; Shi et al. (2025) demonstrate the pattern across K-line generation and volatility forecasting; the slice-and-sum advantage is preserved – an added head, not a replacement.

Counterfactual quantification. Injecting buy or sell orders at 10$\times$ empirical frequency drives rollout mid-price in the expected direction (Appendix G, Figure 9) – a qualitative illustration. Fitting an Almgren–Chriss square-root-law slope β in $\log \Delta p \sim \beta \log Q$ would yield a calibrated market-impact coefficient.

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A. Data, Features, and Mid-Price Estimation

Data and held-out splits. The model is pre-trained on a proprietary dataset built from billions of raw, tick-level US equities transactions, spanning 368 trading days from February 2024 to September 2025, across the breadth of the US equities market. This represents over 19 billion tokens across 1.9 million date-asset pairs. We employ a temporal hold-out strategy, reserving January 2025 onward across all assets for the test set, yielding a training set of 10.7 billion tokens and a test set of 8.7 billion tokens. The tokenizer is calibrated on the first 30 days of the training data, February 2024. For evaluating out-of-distribution generalization we also hold out one month of data from APAC regions, namely Jan. 2025, for both Japan and China.

Mid-Price Estimation. A robust estimate of the true market mid-price ($\hat{p}^{\,\text{mid}}$) is critical for normalizing price-related features. While dedicated market data sources for this exist, they are often expensive. Given our access to raw transaction data, we seek to estimate this value directly. In our partial-information setting, we primarily observe the execution prices ($p^{\,\text{exec}}$) for consummated trades. Fills occur when incoming marketable submissions match resting orders; the modeled event stream carries the submissions and cancellations themselves, while the resulting execution prices are the observable trace used for mid-price estimation. The raw stream of $p^{\,\text{exec}}$ is a noisy version of the true mid-price $\hat{p}^{\,\text{mid}}$.

A naive approach, such as a simple rolling average of execution prices, is insufficient. A fixed-width window of trades is not comparable across assets with different liquidity levels; a 50-trade window may span less than a second for a highly liquid asset but several hours for an illiquid one. A time-based window (e.g., 2 seconds) is more relevant, but a simple average still fails to account for trade volume. For example, an average that gives equal weight to a 1,000-share trade at \$10.00 and a 1-share trade at \$9.00 would produce a misleading estimate.

The conventional solution is the volume-weighted average price (VWAP), which is

$$\hat{p}_t^{\text{VWAP}} = \frac{\sum_{i=0}^{W} v_{t-i}\, p_{t-i}^{\text{exec}}}{\sum_{i=0}^{W} v_{t-i}} \tag{2}$$

To make this estimator more reactive to recent information, we introduce **Exponentially-Weighted Volume-Weighted Average Price (EW-VWAP)**. This is calculated by maintaining two separate exponential moving averages (EMAs): one for the volume-weighted price and one for the volume itself. For each incoming trade with execution price p_t^{exec} and volume v_t, we update the EMAs for the numerator (N_t) and denominator (D_t) as follows:

$$N_t = \alpha \cdot (p_t^{\text{exec}} \cdot v_t) + (1-\alpha) \cdot N_{t-1}$$
$$D_t = \alpha \cdot v_t + (1-\alpha) \cdot D_{t-1}$$

The EW-VWAP at time t is then the ratio of these two values:

$$\hat{p}_t^{\text{EW-VWAP}} = \frac{N_t}{D_t} \tag{3}$$

The smoothing factor α is determined by a time-based halflife, ensuring that the estimate gives more weight to larger and more recent trades in a temporally consistent manner. This provides a stable and representative price benchmark that reflects the price at which the bulk of recent market activity has occurred. This estimator is used throughout the paper to normalize all price-related features to a common scale. Figure 5 compares EW-VWAP against alternative estimators across liquidity profiles.

Scale-Invariant Features. The statistical properties of trading data vary widely across sectors, liquidity profiles, and nominal prices. In raw dollar, share, and second terms, price depths, volumes, and interarrival times for an asset like AAPL may differ greatly from those of a penny stock. Trade representations must therefore be carefully designed to enforce homogeneity across assets. Sirignano & Cont (2019) demonstrate that price formation follows universal principles across stocks: universal models trained on all assets outperform asset-specific models, even for held-out stocks, suggesting deep learning can learn appropriate normalizations from raw data. Building on this, we explicitly construct scale-invariant features to enforce homogeneity, extending universality from order book contexts to heterogeneous event streams across diverse trading venues and market structures.

- **Interarrival Time (Δt_t)**: wall clock time since the previous event: $w_t - w_{t-1}$, in seconds.
- **Log-Transformed Volume (v_t)**: to compress the wide dynamic range of order sizes, which follow heavy-tailed, power-law distributions (Vyetrenko et al., 2019), we apply a logarithmic transformation: $v_t = \log(1 + V_t)$ where V_t is the raw share volume.
- **Normalized Price Depth (δp_t)**: the depth of a limit order with order price p^{order}, relative to the mid-price: $\delta p_t = (p^{\text{order}} - \hat{p}^{\text{mid}})/\hat{p}^{\text{mid}}$. This representation is comparable across differently priced assets, unlike prior work using price depths in ticks.
- **Relative Price Level vs. Open (Δp_t)**: to capture intraday market movement, we quote the current mid-price relative to the day's opening price (p_0): $\Delta p_t = (\hat{p}^{\text{mid}} - p_0)/p_0$.

While price-related features are computed as unit-less ratios, we refer to them in terms of basis points (bps) for interpretability, where a ratio of 0.01 corresponds to 100 bps.

B. Tokenizer Details

Mixed-radix tokenization algorithm. The tokenizer is calibrated once on the first 30 days of the training data and applied per-event during encoding. Continuous features (Δt, δp, v) are digitized via quantile (equal-frequency) bin edges computed on the calibration window after clipping outliers above the 99th and below the 1st percentile (for signed features); discrete features (a, s) are mapped to binary indices. Bin counts are $n_{\Delta t} = n_{\delta p} = n_v = 16$ and $n_a = n_s = 2$.

Algorithm 1 Mixed-radix tokenizer: calibration on the first 30 days of training data, then per-event encoding.

Require: Trade events with features $\{\Delta t, \delta p, v, a, s\}$
Require: Bin counts $n_{\Delta t} = n_{\delta p} = n_v = 16, n_a = n_s = 2$
Calibration phase (once per training corpus):
for each continuous feature $f \in \{\Delta t, \delta p, v\}$ **do**
  Remove NaN and infinite values from f
  Compute upper-tail threshold $u = \text{percentile}(f, 99)$
  if f has signed support **then**
    Compute lower-tail threshold $l = \text{percentile}(f, 1)$
    Clip values outside $[l, u]$ to outlier bins
  else
    Clip values above u to upper outlier bin
  end if
  Compute equal-frequency (quantile) bin edges B_f within the clipped support
end for
Encoding phase (per event):
for each event e **do**
  $I_{\Delta t}, I_{\delta p}, I_v \leftarrow \text{digitize}(e.\Delta t, e.\delta p, e.v)$ using $B_{\Delta t}, B_{\delta p}, B_v$
  $I_a, I_s \leftarrow$ binary indices for action and side
  $T_e \leftarrow I_a \cdot (n_s n_{\delta p} n_v n_{\Delta t}) + I_s \cdot (n_{\delta p} n_v n_{\Delta t}) + I_{\delta p} \cdot (n_v n_{\Delta t}) + I_v \cdot n_{\Delta t} + I_{\Delta t}$
end for
return Token sequence $\{T_e\}$ over vocabulary $|\mathcal{V}| =$ 16,384

Multi-feature token composition. While our model's input at each time step is multi-featured, the decoder is trained to predict a single, unidimensional token, representing the core trade event. Thus, we combine the discrete bin indices of the trade-related features, ($i_{\Delta t}, i_{\delta p}, i_v, i_a, i_s$), into a single composite integer, i_{trade}. This is accomplished by treating the feature indices as digits in a mixed base number system: each feature's bin index is a "digit", and the number of possible values for the subsequent features acts as the "base" at each position. With $n_{\delta p} = 16$ for price depth, $n_v = 16$ for volume, $n_{\Delta t} = 16$ for interarrival time, $n_s = 2$ (buy, sell) for side, and $n_a = 2$ (add or cancel order) for action type, the composite trade token is:

$$
\begin{aligned}
i_{\text{trade}} = &(i_a \times n_s \times n_{\delta p} \times n_v \times n_{\Delta t}) + \\
&(i_s \times n_{\delta p} \times n_v \times n_{\Delta t}) + \\
&(i_{\delta p} \times n_v \times n_{\Delta t}) + (i_v \times n_{\Delta t}) + i_{\Delta t}
\end{aligned} \tag{4}
$$

This yields a vocabulary size of $|\mathcal{V}| = 16{,}384$ for the predictable trade tokens. The model input at each time step is a tuple containing this token along with several non-predicted contextual features used for conditioning. These contextual features are provided as separate inputs, and are not part of the trade token i_{trade} calculated in Equation (4):

- i_l: the liquidity bin index ($n_l = 3$), determined by binning each asset into low, medium, or high liquidity ranges based on its Average Daily Volume (ADV).
- $i_{\Delta p_t}$: the price-level-change bin index ($n_{\Delta p} = 32$).
- I_{MP}: a binary indicator for market-level vs. participant-level sequences.

The final input is $[i_l, I_{MP}, i_{\Delta p_t}, i_{\text{trade}}]$. This formulation allows the model to be conditioned on broader market context while focusing its predictive power on the next trade event.

Worked example. Given the imaginary sequence of trade events in Table 2, our features for timestep $t = 43$ are:

- $\Delta t_t = w_t - w_{t-1} =$ 09:45:38 $-$ 09:45:30 $=$ 8 sec
- $\delta p_t = (p^{\text{order}} - \hat{p}^{\text{mid}})/\hat{p}^{\text{mid}} = (\$182.50 - \$182.48)/\$182.48 = 0.011\% = +1.1$ bps
- $v_t = 750$ shs
- $a_t =$ Add Order
- $s_t =$ Sell

Using our calibrated bins, we discretize to $i_{\Delta t_t} = 11$, $i_{\delta p_t} = 7$, $i_{v_t} = 7$, $i_{a_t} = 0$, $i_{s_t} = 1$. Applying the mixed-radix encoding gives composite token $i_{\text{trade}} = 6{,}011$. Assuming an opening price $p_0 = \$179.50$, the price-level-change feature is $\Delta p_t = 0.017\% = +1.7$ bps, discretizing to $i_{\Delta p_t} = 19$. The asset's average daily volume of 53M places it in the high-liquidity bin ($i_l = 2$). Treating this as a market-level sequence ($I_{MP} = 0$), the final model input is $[i_l, I_{MP}, i_{\Delta p_t}, i_{\text{trade}}] = [2, 0, 19, 6011]$.

Feature stationarity. Figure 6 shows kernel-density estimates of the four scale-invariant features at the tokenizer calibration period (Feb 2024) and one year later (Feb 2025). We observe that our features are empirically stationary across this period even as volatility, price level, and other market conditions vary – the property that licenses tokenizer reuse across the test window.

C. Architecture and Training

Tabular Input Embedding. We employ a tabular embedding approach to handle our multi-feature input tokens. Each feature in the input tuple $[i_l, I_{MP}, i_{\Delta p_t}, i_{\text{trade}}]$ is first projected into its own embedding space using an embedding table. These embedding vectors are concatenated and passed through a linear projection layer to create a unified representation in the model's hidden dimension.

Model Backbone. TradeFM is a decoder-only Transformer, trained from scratch with a custom configuration. The architecture is based on the Llama family (Touvron et al., 2023) and incorporates modern enhancements for efficiency and performance, including:

- **Grouped-Query Attention (GQA)**: Balances the performance of Multi-Head Attention with the reduced memory bandwidth of Multi-Query Attention, enabling faster inference and larger batch sizes.
- **Rotary Position Embeddings (RoPE)**: Encodes relative positional information by applying a rotation to query and key vectors, which has been shown to improve generalization for long sequences.

Model Hyperparameters and Scaling. The model size is guided by the Chinchilla scaling laws, which suggest a compute-optimal ratio of approximately 20 training tokens per model parameter (Kaplan et al., 2020; Hoffmann et al., 2022). Given our dataset of 10.7 billion tokens, this implies a target model size of around 525 million parameters. Our final hyperparameters are as follows:

- **Context Length**: 1,024 tokens
- **Hidden Layers**: 32
- **Embedding Dimension**: 1,024
- **Intermediate MLP Size**: 4,096
- **Attention Heads**: 32
- **Key-Value Heads (GQA)**: 8 heads, 4 groups
- **Total Parameters**: 524.4 Million

The 1:4 ratio between the embedding dimension and the intermediate MLP size is chosen in accordance with best practices for Transformer models (Petty et al., 2024).

Training Configuration. We train the model on an AWS instance with 3 NVIDIA A100 GPUs, each with 80GB of RAM. All training is performed in `fp16` half-precision. To achieve an effective batch size of 4,032, we use a per-device batch size of 24 and gradient accumulation over 56 steps. For further memory optimization and training acceleration, we use the Accelerate library. The model is trained using the AdamW optimizer with a linear learning rate schedule, a learning rate of 5×10^{-5}, and 500 steps of warmup. Following recommendations for training on large datasets (Muennighoff et al., 2023), we train for a

Time-step	Time-stamp	Asset	Avg. Daily Vol. (shs)	Midprice ($)	Action	Side	Order Price ($)	Vol. (shs)
⋮								
42	09:45:30	AAPL	53,496,022	182.45	ADD	BUY	182.44	500
43	09:45:38	AAPL	53,496,022	182.48	ADD	SELL	182.50	750
44	09:45:52	AAPL	53,496,022	182.50	CANCEL	BUY	182.49	300
⋮								

Table 2. **Example trade sequence.** Toy example of trading activity for an imaginary AAPL sequence, demonstrating the multi-feature and heterogeneous nature of our data pre-tokenization.

total of 4 epochs. TFM-ITCH (90M) uses the same architecture family with proportionally smaller width and depth; both variants share the 1,024-token context window and the 16,384-element joint softmax output. A three-scale study in the parent model family (approximately 125M, 250M, and 500M parameters) finds power-law improvement of held-out loss ($\alpha_C \approx 0.18$, $\alpha_D \approx 0.19$) (Kawawa-Beaudan et al., 2026); whether L1/L2 rollout fidelity scales likewise is untested.

D. Simulator Validation

In order to evaluate the simulator, we replay sequences of real orders through it and compare the statistical properties of the resulting simulated trade fills against the real fills from our historical data. We focus on two key metrics: the cumulative distribution function (CDF) of fill volumes and the CDF of lot counts (the number of discrete fills required to complete a single order). We find a strong correspondence between the real and simulated distributions across assets of varying liquidity, confirming that our simulator is a high-fidelity environment for evaluation. We find correlations of 0.91 and 0.98 between simulated and real volumes and lot counts, respectively.

E. KL-Divergence Propagation via Data Processing

We collect here the data-processing inequality that motivates joint event-level training in the body. Let q denote the true conditional distribution over factored events $e \in \mathcal{V}$, and p the model's prediction. The map $\phi : \mathcal{V} \to \mathcal{Y}$ stands in for any of the reductions we care about: slicing the joint into a per-feature marginal, replaying tokens through the matching engine to obtain an L2 snapshot, or collapsing the event stream to a mid-price trajectory. The standard data-processing inequality then gives:

Property E.1 (KL propagation)**.**

$$\mathrm{KL}(\phi_{\#}q \,\|\, \phi_{\#}p) \;\le\; \mathrm{KL}(q \,\|\, p),$$

with equality iff ϕ is invertible q-almost surely on the support of p (Cover & Thomas, 2006, Thm. 2.8.1). The excess NLL at any coarser resolution, $\mathbb{E}_{\phi_{\#}q}[-\log \phi_{\#}p] - H(\phi_{\#}q)$, is therefore at most the excess NLL on the joint.

The same inequality extends to stochastic kernels: $\mathrm{KL}(K_{\#}q \,\|\, K_{\#}p) \le \mathrm{KL}(q \,\|\, p)$ for any Markov kernel K, which covers cases where the downstream map is itself noisy (e.g., stochastic execution).

Scope. The inequality controls KL (and thus excess NLL) on coarser pushforwards. It does not transport calibration in the PIT-uniform sense (Gneiting et al., 2007; Czado et al., 2009), and it gives no direct bound on W_1 or D_n — the headline metrics in Sections 3.2 and 3.3. The connection from joint NLL to L1/L2 distributional fidelity is empirical rather than implied. Randomized PIT on the joint and pushforward auto-calibration are the natural follow-ups (Section 4).

F. Extended L2 Results

As financial markets are dynamic and market regimes are constantly changing, we investigate the tendency of model performance to drift over time. In Figure 7 we extend the aggregated L2 results (Table 1) for all six emergent factors across all nine held-out months: Wasserstein and Kolmogorov-Smirnov distances of generated-vs-real distributions over time, for TradeFM, Compound Hawkes, and the zero-intelligence baseline. We observe that while these metrics vary within a range, the variance is small and our method mostly achieves higher fidelity than baselines. Table 3 provides the per-horizon log-return distances for TFM-ITCH supporting the cross-regime claim in Section 3.3.

Δt (s)	**K-S**			**Wasserstein** ($\times 10^{-4}$)		
	TFM	Hawkes	ZI	TFM	Hawkes	ZI
10	0.066	**0.039**	0.376	**3.32**	4.19	6.38
30	0.130	**0.075**	0.439	**9.35**	12.31	15.21
60	0.190	**0.101**	0.429	**17.55**	23.12	26.61
120	0.226	**0.098**	0.429	**31.49**	43.10	43.23

Table 3. **Cross-regime log-return distributional fidelity for TFM-ITCH (90M).** Per-horizon W_1 ($\times 10^{-4}$) and D_n for the public-ITCH-trained variant evaluated on the 2025 holdout. TFM = TFM-ITCH. Lowest distance per row in bold. TFM-ITCH has the lowest Wasserstein at all four horizons and trails on D_n – the same model-fit pattern as Section 3.2.

Empirical CDFs of log-returns across models. Figure 8 compares the empirical CDFs of mid-price log-returns for real data, TradeFM (TFM-500M), TFM-ITCH, Compound Hawkes, and zero-intelligence. TradeFM and TFM-ITCH track the real ECDF closely; Hawkes shows visible departure (consistent with the K-S distances in Section 3.3); zero-intelligence is nearly a step function. The probability-density view of the same data is in Figure 4 (body).

G. Counterfactual Stress Testing

The integrated TradeFM-simulator system functions as a high-fidelity environment for complex "what-if" analyses and stress testing. This allows for the study of systemic risk and market stability in a controlled environment. The ability to generate plausible, multi-step forecasts of future market trajectories is a direct outcome of this closed-loop simulation capability.

Such systems are also useful to regulators and risk managers (Dwarakanath et al., 2024), who can use this system to simulate the market's response to extreme or counterfactual scenarios, such as by injecting large, anomalous orders into a historical context and observing the resulting price trajectory. Figure 9 demonstrates this capability: for a sample asset, we artificially inject buy or sell orders at $10\times$ the frequency found in the real context, and average mid-price forecasts over 10 rollouts. When we inject artificial sell orders, the mid-price drops, and when we inject buy orders, the mid-price rises, illustrating realistic behavior.

H. Reproducibility

Data. TFM-ITCH is trained on publicly available NASDAQ TotalView-ITCH 5.0 (Jan–Jun 2020), reproducible from public archives. TradeFM (524M) uses a proprietary US equities dataset (Feb 2024 – Sep 2025); the multi-resolution forecasting framework is independent of this dataset, and TFM-ITCH carries the cross-regime evidence (Section 3.3).

Method. The mixed-radix tokenization (Equation (4), Appendix B), EW-VWAP mid-price estimator (Appendix A), scale-invariant feature construction (Appendix A), and slice-and-sum marginal forecasting (Equation (1)) are fully specified in this paper. Full hyperparameter tables and training-curve snapshots are deferred to follow-on work.

Compute. Training TradeFM used 3 NVIDIA A100 GPUs over $\sim$10 days; TFM-ITCH trains on the same hardware in under a day. Inference and evaluation are CPU-feasible.

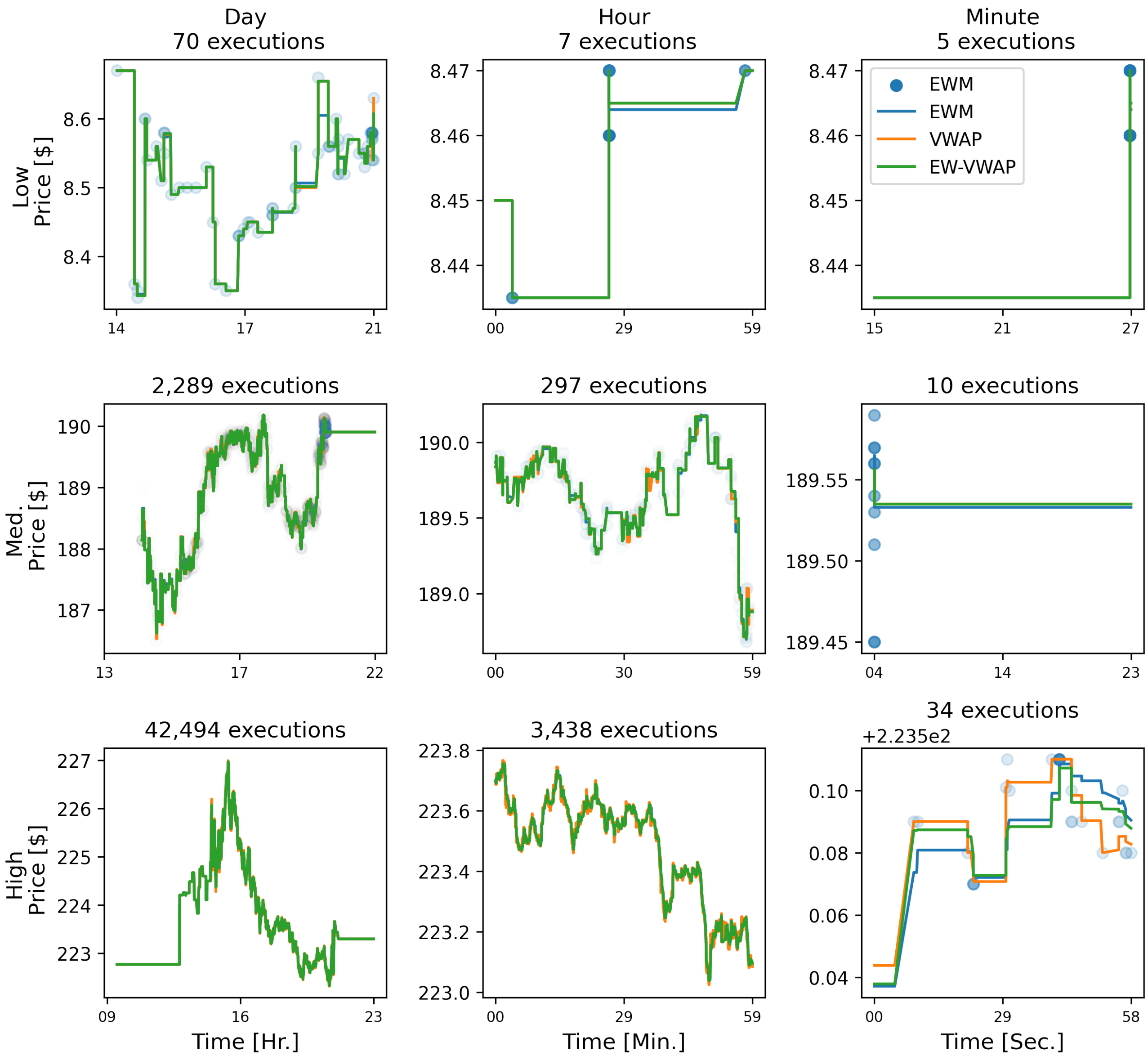

Figure 5. **Mid-price estimator comparison.** EW-VWAP provides a more stable and responsive estimate than standard VWAP or EWM, closely tracking executed fill prices across time scales and liquidity regimes.

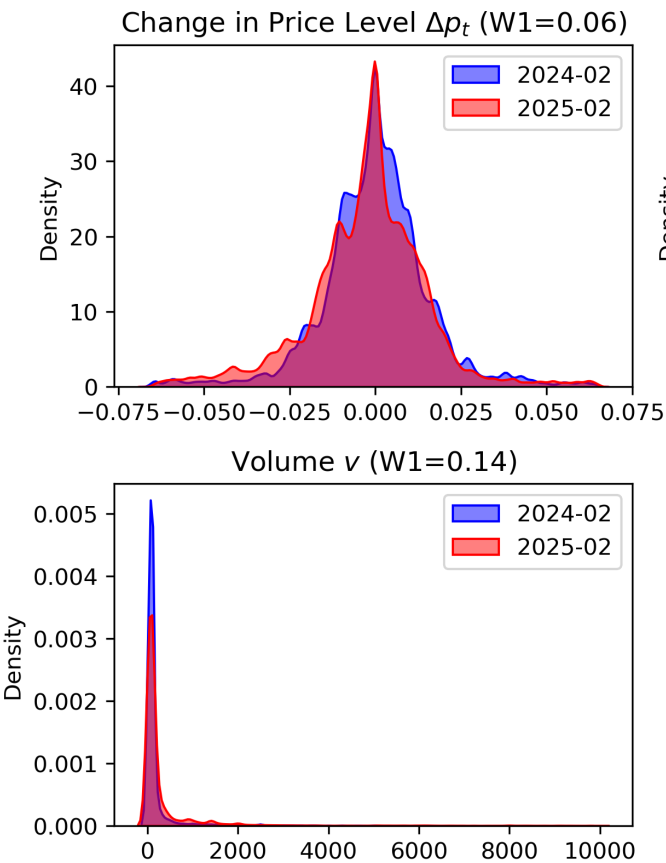

Figure 6. **Feature stationarity.** Kernel-density estimates of scale-invariant features at the tokenizer calibration period (Feb 2024) and one year later (Feb 2025).

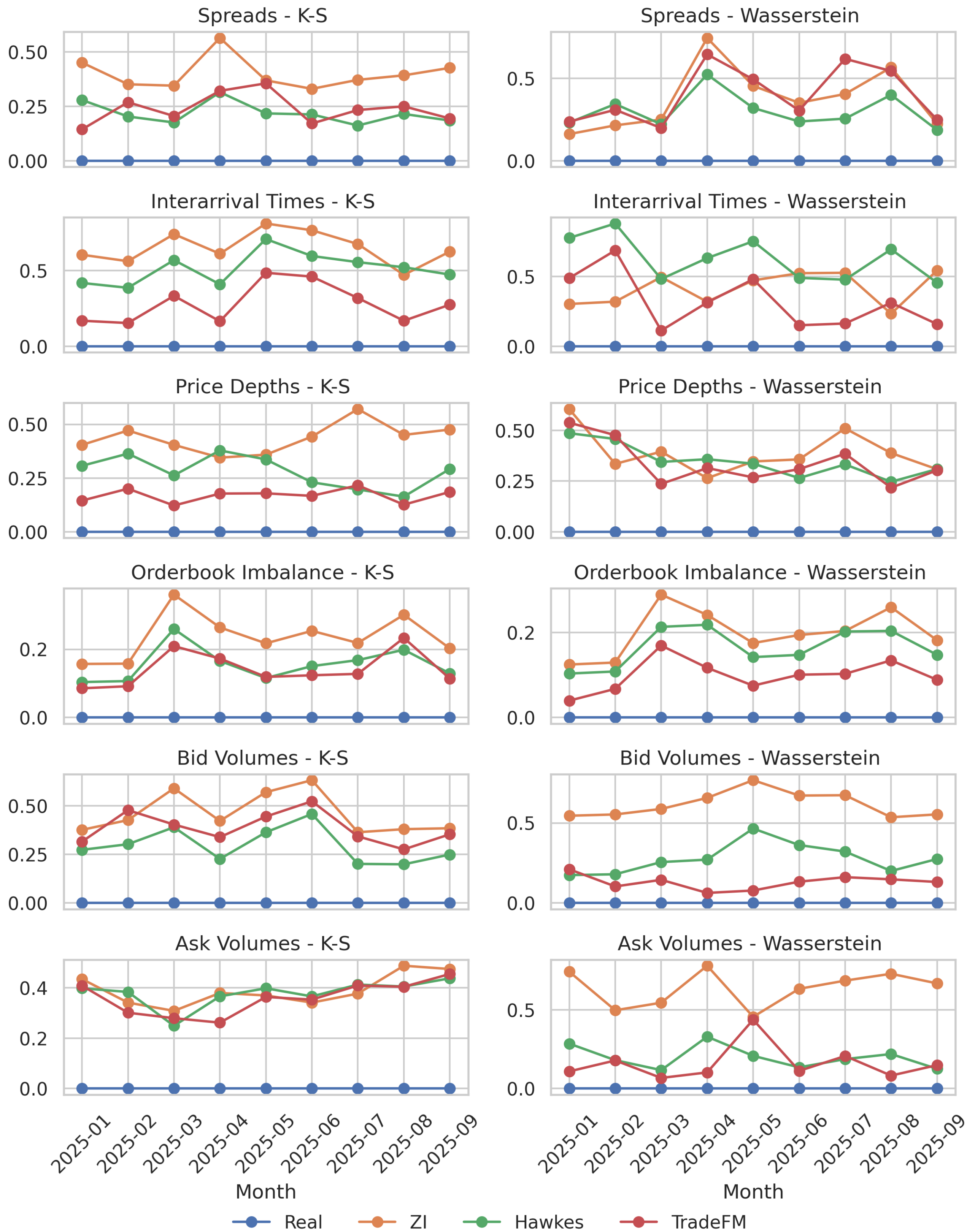

Figure 7. **Distributional fidelity over time.** Wasserstein distance and Kolmogorov-Smirnov statistic of generated-vs-real distributions across emergent market factors, plotted over nine held-out months for TradeFM, Compound Hawkes, and zero-intelligence baseline.

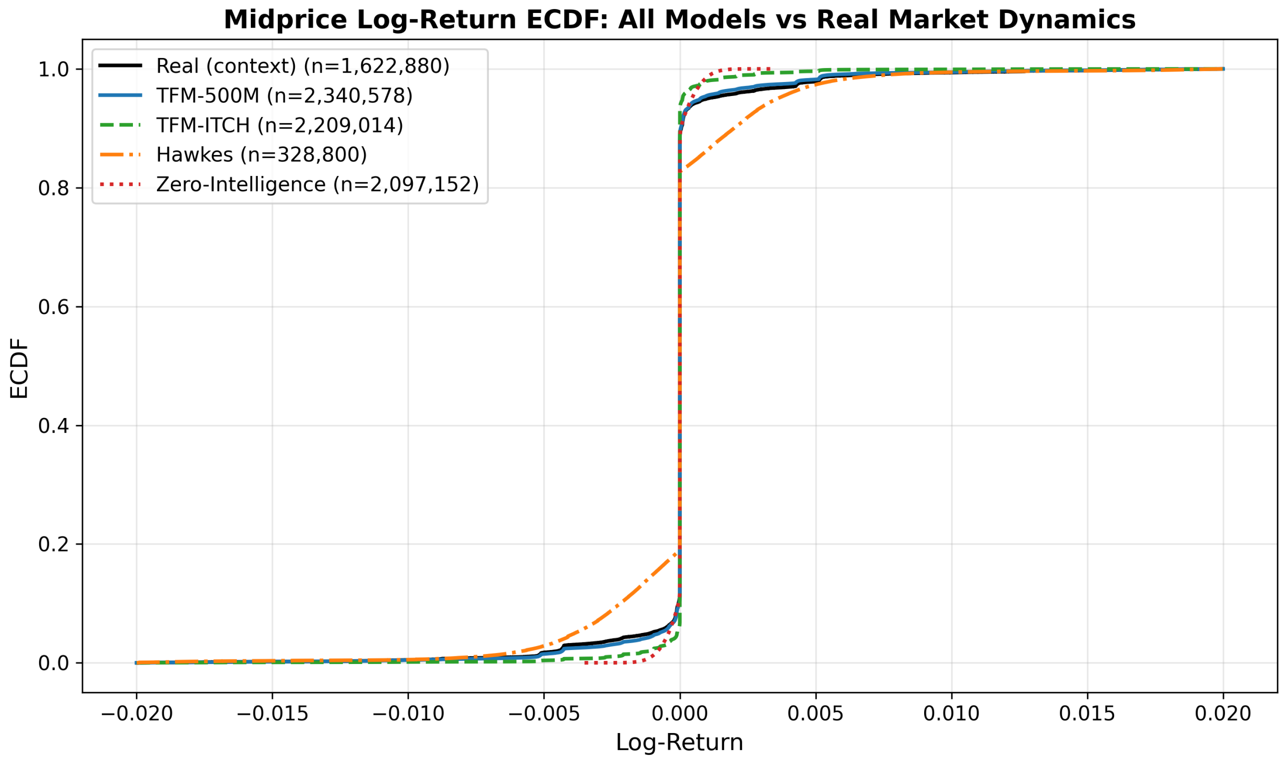

Figure 8. **Mid-price log-return empirical CDFs across models.** TradeFM (TFM-500M) and TFM-ITCH track the real ECDF closely. Hawkes shows visible departure (consistent with the K-S distances in Section 3.3). Zero-intelligence is nearly a step function – it concentrates almost all mass within a narrow window around zero.

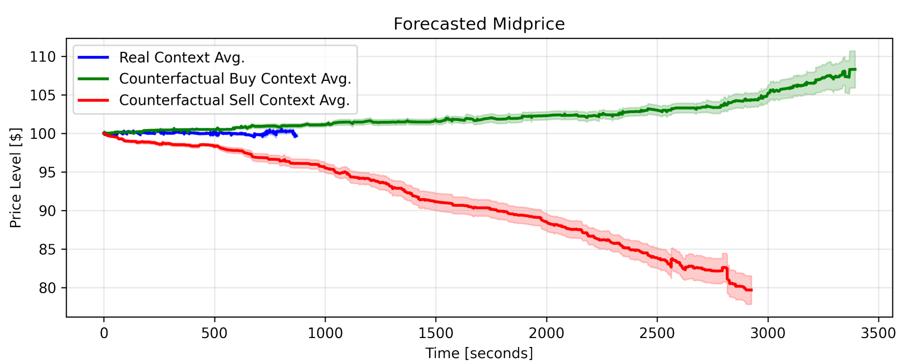

Figure 9. **Counterfactual stress testing.** The model's generated price path responds realistically to injected anomalous order flow (10× normal frequency); qualitative capability, not a calibrated impact measurement.